BRK (00)	Cycles: 7	Size: 2 *
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	** Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2	*** Write to u8(**S**.L+0) | $0100	Push **PC**.H onto stack.
3.1		
3.2	*** Write to u8(**S**.L-1) | $0100	Push **PC**.L onto stack.
4.1		Use Interrupt flags to decide which vector to use. RESET = $FFFC. NMI = $FFFA. IRQ/BRK = $FFFE.
4.2	*** Write to u8(**S**.L-2) | $0100	Push **P** onto stack with **B** flag if software BRK.
5.1		Dec. **S**.L by 3 and set **S**.H to 1.
5.2	Read Vector	Store to Address.L.
6.1		Set **I**, unset **D**.
6.2	Read Vector + 1	Store to Address.H.
7.1		Enable writes to **PC** (disabled by NMI/IRQ). Copy Address to **PC**. Set **PB** to 0.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

* Concerning the BRK instruction, you should also note that although its second byte is basically a “don’t care” byte – that is, it can have any value - the BRK (and COP instruction as well) is a two-byte instruction, the second byte sometimes is used as a signature byte to determine the nature of the BRK being executed. When an RTI instruction is executed, control always returns to the second byte past the BRK opcode. — assembly-programming-manual-for-w65c816.pdf

** If NMI, IRQ, or RESET, OpCode is forced to 0 and writes to PC are disabled.
*** If handling a RESET, the writes turn into reads. The databus is populated but ignored.

ORA (01)	Cycles: 6	Size: 2
	Direct Indexed Indirect (d,x) (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**. Set Pointer to Pointer + **X** + **D**.
		Address.L = Pointer.L.
		Address.H = **D**.H.
2.2		If **D**.L is 0, skip the next cycle.
3.1		Address.H = Pointer.H (fixes high byte of address).
3.2		
4.1		
4.2	Read Address	Store as Pointer.
5.1		
5.2	* If **D**.l is not 0, read (Address + 1), otherwise read (Address & $FF00) | ui8(Address.L + 1)	Store as Pointer.H.
6.1		
6.2	Read Pointer:**DB**	Store as Operand.
7.1		Perform **A** |= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

* The Single-Step Tests require this to be “Read Address + 1,” but this is an error in the tests.

COP (02)	Cycles: 7	Size: 2
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2	Write to u8(**S**.L+0) | $0100	Push **PC**.H onto stack.
3.1		
3.2	Write to u8(**S**.L-1) | $0100	Push **PC**.L onto stack.
4.1		Set Vector to $FFF4.
4.2	Write to u8(**S**.L-2) | $0100	Push **P** onto stack.
5.1		Dec. **S**.L by 3 and set **S**.H to 1.
5.2	Read Vector	Store to Address.L.
6.1		Set **I**, unset **D**.
6.2	Read Vector + 1	Store to Address.H.
7.1		Copy Address to **PC**. Set **PB** to 0.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ORA (03)	Cycles: 4	Size: 2
	Stack Relative (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**. Set Address to Pointer + (**S**.L | $100).
2.2		
3.1		
3.2	Read Address	Store as Operand.
4.1		Perform **A** |= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

TSB (04)	Cycles: 5	Size: 2
	Direct (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.L.
5.1		Set **Z** based off (**A**.L & Operand.L). Perform Operand |= **A**.
5.2	Write to Address	Write Operand.L.
6.1		
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ORA (05)	Cycles: 3	Size: 2
	Direct (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Operand.
4.1		Perform **A** |= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ASL (06)	Cycles: 5	Size: 2
	Direct (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.L.
5.1		Set **C** based off (Operand.L & $80). Perform Operand.L <<= 1. Set **N** based off (Operand.L & $80) and **Z** based off Operand.L.
5.2	Write to Address	Write Operand.L.
6.1		
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ORA (07)	Cycles: 6	Size: 2
	Direct Indirect Long (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Address + 2	Store as Bank.
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform **A** |= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

PHP (08)	Cycles: 3	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		Set Operand.L to **P** with **X** and **M** flags set.
2.2	Write to ((**S**.L | $0100)+0)	Push Operand.L onto stack.
3.1		Dec. **S**.L by 1 and set **S**.H to 1.
3.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ORA (09)	Cycles: 2	Size: 2
	Immediate	
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**. Perform **A** |= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ASL (0A)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Set **C** based off (**A**.L & $80). Perform **A**.L <<= 1. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

PHD (0B)	Cycles: 4	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		
2.2	Write to u8(**S**.L+0) | $0100	Push **D**.H onto stack.
3.1		
3.2	Write to ((**S**.L | $0100)-1)	Push **D**.L onto stack.
4.1		
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

TSB (0C)	Cycles: 6	Size: 3
	Absolute (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address:**DB**	Store as Operand.
4.1		
4.2	Write to Address:**DB**	Write Operand.L.
5.1		Set **Z** based off (**A**.L & Operand.L). Perform Operand |= **A**.
5.2	Write to Address:**DB**	Write Operand.L.
6.1		
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ORA (0D)	Cycles: 4	Size: 3
	Absolute (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address:**DB**	Store as Operand.
4.1		Perform **A** |= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ASL (0E)	Cycles: 6	Size: 3
	Absolute (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address:**DB**	Store as Operand.
4.1		
4.2	Write to Address:**DB**	Write Operand.L.
5.1		Set **C** based off (Operand.L & $80). Perform Operand.L <<= 1. Set **N** based off (Operand.L & $80) and **Z** based off Operand.L.
5.2	Write to Address:**DB**	Write Operand.L.
6.1		
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ORA (0F)	Cycles: 5	Size: 4
	Absolute Long (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read **PC**:**PB**	Stores as Bank.
4.1		Inc. **PC**.
4.2	Read Address:Bank	Store as Operand.
5.1		Perform **A** |= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

BPL (10)	Cycles: 2	Size: 2
	Relative	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**. Decide to branch if **N** is 0.
1.2	Read **PC**:**PB**	Store as Operand. Address = i16(i8(Operand.L)) + **PC**, BoundaryCrossed = Address.H != **PC**.H. If not branching or not BoundaryCrossed, poll interrupts.
2.1		Inc. **PC**. If not branching, end (next half-cycle is 4.2).
2.2		
3.1		Set **PC**.L to Address.L. If not BoundaryCrossed, end (next half-cycle is 4.2).
3.2		Poll interrupts.
4.1		Set **PC**.H to Address.H
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ORA (11)	Cycles: 5	Size: 2
	Direct Indirect Indexed (d),y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Orig = Pointer. Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = **DB** + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform **A** |= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ORA (12)	Cycles: 5	Size: 2
	Direct Indirect (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Pointer:**DB**	Store as Operand.
6.1		Perform **A** |= Operand. Set **N** based off (**A**.L & $80) and Z based off **A**.L.
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ORA (13)	Cycles: 7	Size: 2
	Stack Relative Indirect Indexed (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**.	Store as Pointer.
2.1		Inc. **PC**. Set Address to Pointer + (**S**.L | $100.)
2.2		
3.1		
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). Bank = **DB** + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform **A** |= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
7.2	Read **PC**:**PB**.	Store as OpCode.
+X.1		

TRB (14)	Cycles: 5	Size: 2
	Direct (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.L.
5.1		Set **Z** based off (**A**.L & Operand.L). Perform Operand &= ~**A**.
5.2	Write to Address	Write Operand.L.
6.1		
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ORA (15)	Cycles: 4	Size: 2
	Direct, X (Read)	
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		If **D**.L is 0, Address.L = Operand + **X**.L + **D**.L, Address.H = **D**.H, otherwise Address = Operand + **X** + **D**. If **D**.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1		Perform **A** |= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ASL (16)	Cycles: 6	Size: 2
	Direct, X (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		If **D**.L is 0, Address.L = Operand + **X**.L + **D**.L, Address.H = **D**.H, otherwise Address = Operand + **X** + **D**. If **D**.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1		
5.2	Write to Address	Write Operand.L.
6.1		Set **C** based off (Operand.L & $80). Perform Operand.L <<= 1. Set **N** based off (Operand.L & $80) and **Z** based off Operand.L.
6.2	Write to Address	Write Operand.L.
7.1		
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ORA (17)	Cycles: 6	Size: 2
	Direct Indirect Indexed Long [d],y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	If **D**.l is not 0, read (Address + 1), otherwise read (Address & $FF00) | ui8(Address.L + 1)	Store as Pointer.H.
5.1		
5.2	If **D**.l is not 0, read (Address + 2), otherwise read (Address & $FF00) | ui8(Address.L + 2)	Store as Bank.
6.1		Perform Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). Bank += (Tmp >> 16).
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform **A** |= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

CLC (18)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Set the **C** flag to 0.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ORA (19)	Cycles: 4	Size: 3
	Absolute, Y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**.	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**.	Store as Pointer.H.
3.1		Inc. **PC**. Perform Orig = Pointer. Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = **DB** + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Perform **A** |= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
5.2	Read **PC**:**PB**.	Store as OpCode.
+X.1		

INC (1A)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Perform **A**.L += 1, set **N** based off (**A**.L & $80), set **Z** based off **A**.L.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

TCS (1B)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Set **S**.L to **A**.L and **S**.H to 1.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

TRB (1C)	Cycles: 6	Size: 3
	Absolute (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address:**DB**	Store as Operand.
4.1		
4.2	Write to Address:**DB**	Write Operand.L.
5.1		Set **Z** based off (**A**.L & Operand.L). Perform Operand &= ~**A**.
5.2	Write to Address:**DB**	Write Operand.L.
6.1		
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ORA (1D)	Cycles: 4	Size: 3
	Absolute, X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**.	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**.	Store as Pointer.H.
3.1		Inc. **PC**. Perform Orig = Pointer.
		Tmp = ui32(Pointer + **X**).
		Pointer = ui16(Tmp).
		If Orig.H == Pointer.H, skip 2 half-cycles.
		Bank = **DB** + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Perform **A** |= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
5.2	Read **PC**:**PB**.	Store as OpCode.
+X.1		

ASL (1E)	Cycles: 7	Size: 3
	Absolute, X (Read/Modiy/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**.	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**.	Store as Pointer.H.
3.1		Inc. **PC**. Perform Tmp = ui32(Pointer + **X**). Pointer = ui16(Tmp). Bank = **DB** + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		
5.2	Write to Pointer:Bank	Write Operand.L.
6.1		Set **C** based off (Operand.L & $80). Perform Operand.L <<= 1. Set **N** based off (Operand.L & $80) and **Z** based off Operand.L.
6.2	Write to Pointer:Bank	Write Operand.L.
7.1		
7.2	Read **PC**:**PB**.	Store as OpCode.
+X.1		

ORA (1F)	Cycles: 5	Size: 4
	Absolute Long, X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**.	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**.	Store as Pointer.H.
3.1		Inc. **PC**.
3.2	Read **PC**:**PB**.	Stores as Bank.
4.1		Inc. **PC**. Perform Tmp = ui32(Pointer + **X**). Pointer = ui16(Tmp). Bank += (Tmp >> 16).
4.2	Read Pointer:Bank	Store as Operand.
5.1		Perform **A** |= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
5.2	Read **PC**:**PB**.	Store as OpCode.
+X.1		

JSR (20)	Cycles: 6	Size: 3
	Absolute	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read **PC**:**PB**	Discard.
4.1		Dec. **PC**.
4.2	Write to u8(**S**.L+0) | $0100	Push **PC**.H onto stack.
5.1		
5.2	Write to u8(**S**.L-1) | $0100	Push **PC**.L onto stack.
6.1		Dec. **S**.L by 2 and set **S**.H to 1. Perform **PC** = Address.
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

AND (21)	Cycles: 6	Size: 2
	Direct Indexed Indirect (d,x) (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**. Set Pointer to Pointer + **X** + **D**.
		Address.L = Pointer.L.
		Address.H = **D**.H.
2.2		If **D**.L is 0, skip the next cycle.
3.1		Address.H = Pointer.H (fixes high byte of address).
3.2		
4.1		
4.2	Read Address	Store as Pointer.
5.1		
5.2	* If **D**.l is not 0, read (Address + 1), otherwise read (Address & $FF00) | ui8(Address.L + 1)	Store as Pointer.H.
6.1		
6.2	Read Pointer:**DB**	Store as Operand.
7.1		Perform **A** &= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

* The Single-Step Tests require this to be “Read Address + 1,” but this is an error in the tests.

JSL (22)	Cycles: 8	Size: 4
	Absolute Long	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Write to u8(**S**.L+0) | $0100	Push **PB** onto stack.
4.1		
4.2	Read u8(**S**.L+0) | $0100	Discard.
5.1		
5.2	Read **PC**:**PB**	Stores as Bank.
6.1		
6.2	Write to ((**S**.L | $0100)-1)	Push **PC**.H onto stack.
7.1		
7.2	Write to ((**S**.L | $0100)-2)	Push **PC**.L onto stack.
8.1		Dec. **S**.L by 3 and set **S**.H to 1. Copy Address to **PC**. Copy Bank to **PB**.
8.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

AND (23)	Cycles: 4	Size: 2
	Stack Relative (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**.	Store as Pointer.
2.1		Inc. **PC**. Set Address to Pointer + (**S**.L | $0100).
2.2		
3.1		
3.2	Read Address	Store as Operand.
4.1		Perform **A** &= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
4.2	Read **PC**:**PB**.	Store as OpCode.
+X.1		

BIT (24)	Cycles: 3	Size: 2
	Direct (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Operand.
4.1		Set **Z** based off (**A**.L & Operand.L), set **N** based off (Operand.L & $80), and set **V** based off (Operand.L & $40).
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

AND (25)	Cycles: 3	Size: 2
	Direct (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Operand.
4.1		Perform **A** &= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ROL (26)	Cycles: 5	Size: 2
	Direct (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.L.
5.1		Set **C** based off (Operand.L & $80). Perform Operand.L = (Operand.L << 1) | **C**. Set **N** based off (Operand.L & $80) and **Z** based off Operand.L.
5.2	Write to Address	Write Operand.L.
6.1		
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

AND (27)	Cycles: 6	Size: 2
	Direct Indirect Long (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Address + 2	Store as Bank.
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform **A** &= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

PLP (28)	Cycles: 4	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		
2.2		
3.1		Inc. **S**.L by 1 and set **S**.H to 1.
3.2	Read **S**.L | $0100	Store as Operand.L.
4.1		**P** = Operand.L, set **X** and **M** flags, set **X**.H and **Y**.H to 0.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

AND (29)	Cycles: 2	Size: 2
	Immediate	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**. Perform **A** &= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ROL (2A)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Set **C** based off (**A**.L & $80). Perform **A**.L = (**A**.L << 1) | **C**. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

PLD (2B)	Cycles: 5	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		
2.2		
3.1		
3.2	Read ((**S**.L | $0100)+1)	Store as Operand.L.
4.1		
4.2	Read ((**S**.L | $0100)+2)	Store as Operand.H.
5.1		Inc. **S**.L by 2 and set **S**.H to 1. Set **D** to Operand. Set **N** based off (**D**.H & $80). Set **Z** based off **D**.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

BIT (2C)	Cycles: 4	Size: 3
	Absolute (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address:**DB**	Store as Operand.
4.1		Set **Z** based off (**A**.L & Operand.L), set **N** based off (Operand.L & $80), and set **V** based off (Operand.L & $40).
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

AND (2D)	Cycles: 4	Size: 3
	Absolute (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address:**DB**	Store as Operand.
4.1		Perform **A** &= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ROL (2E)	Cycles: 6	Size: 3
	Absolute (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address:**DB**	Store as Operand.
4.1		
4.2	Write to Address:**DB**	Write Operand.L.
5.1		Set **C** based off (Operand.L & $80). Perform Operand.L = (Operand.L << 1) | **C**. Set **N** based off (Operand.L & $80) and **Z** based off Operand.L.
5.2	Write to Address:**DB**	Write Operand.L.
6.1		
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

AND (2F)	Cycles: 5	Size: 4
	Absolute Long (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		**Inc. PC.**
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		**Inc. PC.**
3.2	Read **PC**:**PB**	Stores as Bank.
4.1		**Inc. PC.**
4.2	Read Address:Bank	Store as Operand.
5.1		Perform **A** &= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

BMI (30)	Cycles: 2	Size: 2
	Relative	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**. Decide to branch if **N** is 1.
1.2	Read **PC**:**PB**	Store as Operand. Address = i16(i8(Operand.L)) + **PC**, BoundaryCrossed = Address.H != **PC**.H. If not branching or not BoundaryCrossed, poll interrupts.
2.1		Inc. **PC**. If not branching, end (next half-cycle is 4.2).
2.2		
3.1		Set **PC**.L to Address.L. If not BoundaryCrossed, end (next half-cycle is 4.2).
3.2		Poll interrupts.
4.1		Set **PC**.H to Address.H
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

AND (31)	Cycles: 5	Size: 2
	Direct Indirect Indexed (d),y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Orig = Pointer. Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = **DB** + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform **A** &= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

AND (32)	Cycles: 5	Size: 2
	Direct Indirect (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Pointer:**DB**	Store as Operand.
6.1		Perform **A** &= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

AND (33)	Cycles: 7	Size: 2
	Stack Relative Indirect Indexed (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**.	Store as Pointer.
2.1		Inc. **PC**. Set Address to Pointer + (**S**.L | $0100).
2.2		
3.1		
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). Bank = **DB** + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform **A** &= Operand. Set **N** based off (**A.**L & $80) and **Z** based off **A.**L.
7.2	Read **PC**:**PB**.	Store as OpCode.
+X.1		

BIT (34)	Cycles: 4	Size: 2
	Direct, X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**.	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		If **D**.L is 0, Address.L = Operand + **X**.L + **D**.L, Address.H = **D**.H, otherwise Address = Operand + **X** + **D**. If **D**.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1		Set **Z** based off (**A**.L & Operand.L), set **N** based off (Operand.L & $80), and set **V** based off (Operand.L & $40).
5.2	Read **PC**:**PB**.	Store as OpCode.
+X.1		

AND (35)	Cycles: 4	Size: 2
	Direct, X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		If **D**.L is 0, Address.L = Operand + **X**.L + **D**.L, Address.H = **D**.H, otherwise Address = Operand + **X** + **D**. If **D**.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1		Perform **A** &= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ROL (36)	Cycles: 6	Size: 2
	Direct, X (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		If **D**.L is 0, Address.L = Operand + **X**.L + **D**.L, Address.H = **D**.H, otherwise Address = Operand + **X** + **D**. If **D**.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1		
5.2	Write to Address	Write Operand.L.
6.1		Set **C** based off (Operand.L & $80). Perform Operand.L = (Operand.L << 1) | **C**. Set **N** based off (Operand.L & $80) and **Z** based off Operand.L.
6.2	Write to Address	Write Operand.L.
7.1		
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

AND (37)	Cycles: 6	Size: 2
	Direct Indirect Indexed Long [d],y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	If **D**.l is not 0, read (Address + 1), otherwise read (Address & $FF00) | ui8(Address.L + 1)	Store as Pointer.H.
5.1		
5.2	If **D**.l is not 0, read (Address + 2), otherwise read (Address & $FF00) | ui8(Address.L + 2)	Store as Bank.
6.1		Perform Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). Bank += (Tmp >> 16).
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform **A** &= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

SEC (38)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Set the **C** flag to 1.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

AND (39)	Cycles: 4	Size: 3
	Absolute, Y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**. Perform Orig = Pointer. Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = **DB** + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Perform **A** &= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

DEC (3A)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Perform **A**.L -= 1, set **N** based off (**A**.L & $80), set **Z** based off **A**.L.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

TSC (3B)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Set **A** to (**S**.L | $0100). Set **N** based off (**A**.H & $80) and **Z** based off **A**.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

BIT (3C)	Cycles: 4	Size: 3
	Absolute, X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**. Perform Orig = Pointer.
		Tmp = ui32(Pointer + **X**).
		Pointer = ui16(Tmp).
		If Orig.H == Pointer.H, skip 2 half-cycles.
		Bank = **DB** + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Set **Z** based off (**A**.L & Operand.L), set **N** based off (Operand.L & $80), and set **V** based off (Operand.L & $40).
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

AND (3D)	Cycles: 4	Size: 3
	Absolute, X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**. Perform Orig = Pointer.
		Tmp = ui32(Pointer + **X**).
		Pointer = ui16(Tmp).
		If Orig.H == Pointer.H, skip 2 half-cycles.
		Bank = **DB** + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Perform **A** &= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ROL (3E)	Cycles: 7	Size: 3
	Absolute, X (Read/Modiy/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**.	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**.	Store as Pointer.H.
3.1		Inc. **PC**. Perform Tmp = ui32(Pointer + **X**). Pointer = ui16(Tmp). Bank = **DB** + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		
5.2	Write to Pointer:Bank	Write Operand.L.
6.1		Set **C** based off (Operand.L & $80). Perform Operand.L = (Operand.L << 1) | **C**. Set **N** based off (Operand.L & $80) and **Z** based off Operand.L.
6.2	Write to Pointer:Bank	Write Operand.L.
7.1		
7.2	Read **PC**:**PB**.	Store as OpCode.
+X.1		

AND (3F)	Cycles: 5	Size: 4
	Absolute Long, X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**.	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**.	Store as Pointer.H.
3.1		Inc. **PC**.
3.2	Read **PC**:**PB**.	Stores as Bank.
4.1		Inc. **PC**. Perform Tmp = ui32(Pointer + **X**). Pointer = ui16(Tmp). Bank += (Tmp >> 16).
4.2	Read Pointer:Bank	Store as Operand.
5.1		Perform **A** &= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
5.2	Read **PC**:**PB**.	Store as OpCode.
+X.1		

RTI (40)	Cycles: 6	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		
2.2		
3.1		Inc. **S**.L by 1 and set **S**.H to 1.
3.2	Read ((**S**.L | $0100)+0)	Store as Operand.L.
4.1		Set **P** to ((Operand.L & ~$30) | (**P** & $30)).
4.2	Read ((**S**.L | $0100)+1)	Store as Operand.L.
5.1		
5.2	Read ((**S**.L | $0100)+2)	Store as Operand.H.
6.1		Inc. **S**.L by 2 and set **S**.H to 1. Set **PC** to Operand.
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

EOR (41)	Cycles: 6	Size: 2
	Direct Indexed Indirect (d,x) (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**. Set Pointer to Pointer + **X** + **D**.
		Address.L = Pointer.L.
		Address.H = **D**.H.
2.2		If **D**.L is 0, skip the next cycle.
3.1		Address.H = Pointer.H (fixes high byte of address).
3.2		
4.1		
4.2	Read Address	Store as Pointer.
5.1		
5.2	* If **D**.l is not 0, read (Address + 1), otherwise read (Address & $FF00) | ui8(Address.L + 1)	Store as Pointer.H.
6.1		
6.2	Read Pointer:**DB**	Store as Operand.
7.1		Perform **A** ^= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

* The Single-Step Tests require this to be “Read Address + 1,” but this is an error in the tests.

WDM (42)	Cycles: 2	Size: 2
	Immediate	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

EOR (43)	Cycles: 4	Size: 2
	Stack Relative (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**. Set Address to Pointer + (**S**.L | $0100).
2.2		
3.1		
3.2	Read Address	Store as Operand.
4.1		Perform **A** ^= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

MVP (44)	Cycles: 7	Size: 3
	Block Move (Positive)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**. Set **DB** to Operand.L.
2.2	Read **PC**:**PB**	Stores as Bank.
3.1		Inc. **PC**.
3.2	Read **X**:Bank	Store as Operand.L.
4.1		
4.2	Write **Y**:**DB**	Write Operand.L.
5.1		Perform **A** -= 1. If **X** flag is set, perform **X**.L -= 1 and **Y**.L -= 1, otherwise perform **X** -= 1 and **Y** -= 1. If **A** is not $FFFF, perform **PC** -= 3.
5.2		
6.1		
6.2		
7.1		
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

EOR (45)	Cycles: 3	Size: 2
	Direct (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Operand.
4.1		Perform **A** ^= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LSR (46)	Cycles: 5	Size: 2
	Direct (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.L.
5.1		Set **C** based off (Operand.L & $01). Perform Operand.L >>= 1. Set **N** based off (Operand.L & $80) and **Z** based off Operand.L.
5.2	Write to Address	Write Operand.L.
6.1		
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

EOR (47)	Cycles: 6	Size: 2
	Direct Indirect Long (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Address + 2	Store as Bank.
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform **A** ^= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

PHA (48)	Cycles: 3	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		
2.2	Write to ((**S**.L | $0100)+0)	Push **A**.L onto stack.
3.1		Dec. **S**.L by 1 and set **S**.H to 1.
3.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

EOR (49)	Cycles: 2	Size: 2
	Immediate	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**. Perform **A** ^= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LSR (4A)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Set **C** based off (**A**.L & $01). Perform **A**.L >>= 1. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

PHK (4B)	Cycles: 3	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		
2.2	Write to ((**S**.L | $0100)+0)	Push **PB** onto stack.
3.1		Dec. **S**.L by 1 and set **S**.H to 1.
3.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

JMP (4C)	Cycles: 3	Size: 3
	Absolute	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Set **PC** to Address.
3.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

EOR (4D)	Cycles: 4	Size: 3
	Absolute (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address:**DB**	Store as Operand.
4.1		Perform **A** ^= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LSR (4E)	Cycles: 6	Size: 3
	Absolute (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address:**DB**	Store as Operand.
4.1		
4.2	Write to Address:**DB**	Write Operand.L.
5.1		Set **C** based off (Operand.L & $01). Perform Operand.L >>= 1. Set **N** based off (Operand.L & $80) and **Z** based off Operand.L.
5.2	Write to Address:**DB**	Write Operand.L.
6.1		
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

EOR (4F)	Cycles: 5	Size: 4
	Absolute Long (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read **PC**:**PB**	Stores as Bank.
4.1		Inc. **PC**.
4.2	Read Address:Bank	Store as Operand.
5.1		Perform **A** ^= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

BVC (50)	Cycles: 2	Size: 2
	Relative	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**. Decide to branch if **V** is 0.
1.2	Read **PC**:**PB**	Store as Operand. Address = i16(i8(Operand.L)) + **PC**, BoundaryCrossed = Address.H != **PC**.H. If not branching or not BoundaryCrossed, poll interrupts.
2.1		Inc. **PC**. If not branching, end (next half-cycle is 4.2).
2.2		
3.1		Set **PC**.L to Address.L. If not BoundaryCrossed, end (next half-cycle is 4.2).
3.2		Poll interrupts.
4.1		Set **PC**.H to Address.H
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

EOR (51)	Cycles: 5	Size: 2
	Direct Indirect Indexed (d),y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Orig = Pointer. Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = **DB** + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform **A** ^= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

EOR (52)	Cycles: 5	Size: 2
	Direct Indirect (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Pointer:**DB**	Store as Operand.
6.1		Perform **A** ^= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

EOR (53)	Cycles: 7	Size: 2
	Stack Relative Indirect Indexed (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**. Set Address to Pointer + (**S**.L | $0100).
2.2		
3.1		
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). Bank = **DB** + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform **A** ^= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

MVN (54)	Cycles: 7	Size: 3
	Block Move (Negative)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**. Set **DB** to Operand.L.
2.2	Read **PC**:**PB**	Stores as Bank.
3.1		Inc. **PC**.
3.2	Read **X**:Bank	Store as Operand.L.
4.1		
4.2	Write **Y**:**DB**	Write Operand.L.
5.1		Perform **A** -= 1. If **X** flag is set, perform **X**.L += 1 and **Y**.L += 1, otherwise perform **X** += 1 and **Y** += 1. If **A** is not $FFFF, perform **PC** -= 3.
5.2		
6.1		
6.2		
7.1		
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

EOR (55)	Cycles: 4	Size: 2
	Direct, X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		If **D**.L is 0, Address.L = Operand + **X**.L + **D**.L, Address.H = **D**.H, otherwise Address = Operand + **X** + **D**. If **D**.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1		Perform **A** ^= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LSR (56)	Cycles: 6	Size: 2
	Direct, X (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		If **D**.L is 0, Address.L = Operand + **X**.L + **D**.L, Address.H = **D**.H, otherwise Address = Operand + **X** + **D**. If **D**.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1		
5.2	Write to Address	Write Operand.L.
6.1		Set **C** based off (Operand.L & $01). Perform Operand.L >>= 1. Set **N** based off (Operand.L & $80) and **Z** based off Operand.L.
6.2	Write to Address	Write Operand.L.
7.1		
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

EOR (57)	Cycles: 6	Size: 2
	Direct Indirect Indexed Long [d],y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	If **D**.l is not 0, read (Address + 1), otherwise read (Address & $FF00) | ui8(Address.L + 1)	Store as Pointer.H.
5.1		
5.2	If **D**.l is not 0, read (Address + 2), otherwise read (Address & $FF00) | ui8(Address.L + 2)	Store as Bank.
6.1		Perform Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). Bank += (Tmp >> 16).
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform **A** ^= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

CLI (58)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Set the **I** flag to 0.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

EOR (59)	Cycles: 4	Size: 3
	Absolute, Y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**. Perform Orig = Pointer.
		Tmp = ui32(Pointer + **Y**).
		Pointer = ui16(Tmp).
		If Orig.H == Pointer.H, skip 2 half-cycles.
		Bank = **DB** + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Perform **A** ^= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

PHY (5A)	Cycles: 3	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		
2.2	Write to ((**S**.L | $0100)+0)	Push **Y**.L onto stack.
3.1		Dec. **S**.L by 1 and set **S**.H to 1.
3.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

TCD (5B)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Set **D** to **A**. Set **N** based off (**D**.H & $80) and **Z** based off **D**.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

JML (5C)	Cycles: 4	Size: 4
	Absolute Long	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read **PC**:**PB**	Stores as Bank.
4.1		Copy Address to **PC**. Copy Bank to **PB**.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

EOR (5D)	Cycles: 4	Size: 3
	Absolute, X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**.	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**.	Store as Pointer.H.
3.1		Inc. **PC**. Perform Orig = Pointer. Tmp = ui32(Pointer + **X**). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = **DB** + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Perform **A** ^= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LSR (5E)	Cycles: 7	Size: 3
	Absolute, X (Read/Modiy/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**. Perform Tmp = ui32(Pointer + **X**). Pointer = ui16(Tmp). Bank = **DB** + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		
5.2	Write to Pointer:Bank	Write Operand.L.
6.1		Set **C** based off (Operand.L & $01). Perform Operand.L >>= 1. Set **N** based off (Operand.L & $80) and **Z** based off Operand.L.
6.2	Write to Pointer:Bank	Write Operand.L.
7.1		
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

EOR (5F)	Cycles: 5	Size: 4
	Absolute Long, X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**.
3.2	Read **PC**:**PB**	Stores as Bank.
4.1		Inc. **PC**. Perform Tmp = ui32(Pointer + **X**).
		Pointer = ui16(Tmp).
		Bank += (Tmp >> 16).
4.2	Read Pointer:Bank	Store as Operand.
5.1		Perform **A** ^= Operand. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

RTS (60)	Cycles: 6	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		
2.2		
3.1		Inc. **S**.L by 1 and set **S**.H to 1.
3.2	Read ((**S**.L | $0100)+0)	Store as Operand.L.
4.1		
4.2	Read ((**S**.L | $0100)+1)	Store as Operand.H.
5.1		
5.2		
6.1		Inc. **S**.L by 1 and set **S**.H to 1. Set **PC** to (Operand + 1).
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ADC (61)	Cycles: 6	Size: 2
	Direct Indexed Indirect (d,x) (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**. Set Pointer to Pointer + **X** + **D**.
		Address.L = Pointer.L.
		Address.H = **D**.H.
2.2		If **D**.L is 0, skip the next cycle.
3.1		Address.H = Pointer.H (fixes high byte of address).
3.2		
4.1		
4.2	Read Address	Store as Pointer.
5.1		
5.2	* If **D**.l is not 0, read (Address + 1), otherwise read (Address & $FF00) | ui8(Address.L + 1)	Store as Pointer.H.
6.1		
6.2	Read Pointer:**DB**	Store as Operand.

7.1		If **D** flag == 0:
		Result = ui16(**A**.L) + ui16(Operand) + **C**.
		V flag = (~(ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result > $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = ui16(**A**.L & 0x0F) + ui16(Operand & 0x0F) + **C**;
		if (Lo > 9) Lo += 6.
		CarryToHi = Lo > $0F ? 1 : 0.
		HiSum = ui16(**A**.L >> 4) + ui16(Operand >> 4) + CarryToHi.
		V flag = (~((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum > 9) HiSum += 6.
		C flag = HiSum > $0F.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

* The Single-Step Tests require this to be “Read Address + 1,” but this is an error in the tests.

PER (62)	Cycles: 6	Size: 3
	Program Counter Relative Long	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Operand.H.
3.1		Inc. **PC**. Set Operand to Operand + **PC**.
3.2		
4.1		
4.2	Write to ((**S**.L | $0100)+0)	Push Operand.H onto stack.
5.1		
5.2	Write to ((**S**.L | $0100)-1)	Push Operand.L onto stack.
6.1		Dec. **S**.L by 2 and set **S**.H to 1.
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ADC (63)	Cycles: 4	Size: 2
	Stack Relative (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**. Set Address to Pointer + (**S**.L | $0100).
2.2		
3.1		
3.2	Read Address	Store as Operand.

4.1		If **D** flag == 0:
		Result = ui16(**A**.L) + ui16(Operand) + **C**.
		V flag = (~(ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result > $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = ui16(**A**.L & 0x0F) + ui16(Operand & 0x0F) + **C**;
		if (Lo > 9) Lo += 6.
		CarryToHi = Lo > $0F ? 1 : 0.
		HiSum = ui16(**A**.L >> 4) + ui16(Operand >> 4) + CarryToHi.
		V flag = (~((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum > 9) HiSum += 6.
		C flag = HiSum > $0F.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STZ (64)	Cycles: 3	Size: 2
	Direct (Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Write to Address	Write 0.
4.1		
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ADC (65)	Cycles: 3	Size: 2
	Direct (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Operand.

4.1		If **D** flag == 0:
		Result = ui16(**A**.L) + ui16(Operand) + **C**.
		V flag = (~(ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result > $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = ui16(**A**.L & 0x0F) + ui16(Operand & 0x0F) + **C**;
		if (Lo > 9) Lo += 6.
		CarryToHi = Lo > $0F ? 1 : 0.
		HiSum = ui16(**A**.L >> 4) + ui16(Operand >> 4) + CarryToHi.
		V flag = (~((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum > 9) HiSum += 6.
		C flag = HiSum > $0F.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ROR (66)	Cycles: 5	Size: 2
	Direct (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.L.
5.1		Set **C** based off (Operand.L & $01). Perform Operand.L = (Operand.L >> 1) | (**C** << 7). Set **N** based off (Operand.L & $80) and **Z** based off Operand.L.
5.2	Write to Address	Write Operand.L.
6.1		
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ADC (67)	Cycles: 6	Size: 2
	Direct Indirect Long (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Address + 2	Store as Bank.
6.1		
6.2	Read Pointer:Bank	Store as Operand.

7.1		If **D** flag == 0:
		Result = ui16(**A**.L) + ui16(Operand) + **C**.
		V flag = (~(ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result > $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = ui16(**A**.L & 0x0F) + ui16(Operand & 0x0F) + **C**;
		if (Lo > 9) Lo += 6.
		CarryToHi = Lo > $0F ? 1 : 0.
		HiSum = ui16(**A**.L >> 4) + ui16(Operand >> 4) + CarryToHi.
		V flag = (~((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum > 9) HiSum += 6.
		C flag = HiSum > $0F.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

PLA (68)	Cycles: 4	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		
2.2		
3.1		Inc. **S**.L by 1 and set **S**.H to 1.
3.2	Read u8(**S**.L+0) | $0100	Store as Operand.L.
4.1		Set **A**.L to Operand.L. Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ADC (69)	Cycles: 2	Size: 2
	Immediate	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.

2.1		If **D** flag == 0:
		Result = ui16(**A**.L) + ui16(Operand) + **C**.
		V flag = (~(ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result > $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = ui16(**A**.L & 0x0F) + ui16(Operand & 0x0F) + **C**;
		if (Lo > 9) Lo += 6.
		CarryToHi = Lo > $0F ? 1 : 0.
		HiSum = ui16(**A**.L >> 4) + ui16(Operand >> 4) + CarryToHi.
		V flag = (~((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum > 9) HiSum += 6.
		C flag = HiSum > $0F.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ROR (6A)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Set **C** based off (**A**.L & $01). Perform **A**.L = (**A**.L >> 1) | (**C** << 7). Set **N** based off (**A**.L & $80) and **Z** based off **A**.L.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

RTL (6B)	Cycles: 6	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		
2.2		
3.1		
3.2	Read u8(**S**.L+1) | $0100	Store as Operand.L.
4.1		
4.2	Read u8(**S**.L+2) | $0100	Store as Operand.H.
5.1		
5.2	Read ((**S**.L | $0100)+3)	Store as **PB**.
6.1		Inc. **S**.L by 3 and set **S**.H to 1. Set **PC** to (Operand + 1). Set **PB** to Bank.
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

JMP (6C)	Cycles: 5	Size: 3
	Absolute Indirect (Jump)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**.
3.2	Read Pointer	Store as Address.
4.1		
4.2	Read Pointer + 1	Store as Address.H.
5.1		Set **PC** to Address.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ADC (6D)	Cycles: 4	Size: 3
	Absolute (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address:**DB**	Store as Operand.

4.1		If **D** flag == 0:
		Result = ui16(**A**.L) + ui16(Operand) + **C**.
		V flag = (~(ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result > $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = ui16(**A**.L & 0x0F) + ui16(Operand & 0x0F) + **C**;
		if (Lo > 9) Lo += 6.
		CarryToHi = Lo > $0F ? 1 : 0.
		HiSum = ui16(**A**.L >> 4) + ui16(Operand >> 4) + CarryToHi.
		V flag = (~((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum > 9) HiSum += 6.
		C flag = HiSum > $0F.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ROR (6E)	Cycles: 6	Size: 3
	Absolute (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address:**DB**	Store as Operand.
4.1		
4.2	Write to Address:**DB**	Write Operand.L.
5.1		Set **C** based off (Operand.L & $01). Perform Operand.L = (Operand.L >> 1) | (**C** << 7). Set **N** based off (Operand.L & $80) and **Z** based off Operand.L.
5.2	Write to Address:**DB**	Write Operand.L.
6.1		
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ADC (6F)	Cycles: 5	Size: 4
	Absolute Long (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read **PC**:**PB**	Stores as Bank.
4.1		Inc. **PC**.
4.2	Read Address:Bank	Store as Operand.

5.1		If **D** flag == 0:
		Result = ui16(**A**.L) + ui16(Operand) + **C**.
		V flag = (~(ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result > $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = ui16(**A**.L & 0x0F) + ui16(Operand & 0x0F) + **C**;
		if (Lo > 9) Lo += 6.
		CarryToHi = Lo > $0F ? 1 : 0.
		HiSum = ui16(**A**.L >> 4) + ui16(Operand >> 4) + CarryToHi.
		V flag = (~((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum > 9) HiSum += 6.
		C flag = HiSum > $0F.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

BVS (70)	Cycles: 2	Size: 2
	Relative	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**. Decide to branch if **V** is 1.
1.2	Read **PC**:**PB**	Store as Operand. Address = i16(i8(Operand.L)) + **PC**, BoundaryCrossed = Address.H != **PC**.H. If not branching or not BoundaryCrossed, poll interrupts.
2.1		Inc. **PC**. If not branching, end (next half-cycle is 4.2).
2.2		
3.1		Set **PC**.L to Address.L. If not BoundaryCrossed, end (next half-cycle is 4.2).
3.2		Poll interrupts.
4.1		Set **PC**.H to Address.H
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ADC (71)	Cycles: 5	Size: 2
	Direct Indirect Indexed (d),y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Orig = Pointer. Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = **DB** + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.

7.1		If **D** flag == 0:
		Result = ui16(**A**.L) + ui16(Operand) + **C**.
		V flag = (~(ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result > $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = ui16(**A**.L & 0x0F) + ui16(Operand & 0x0F) + **C**;
		if (Lo > 9) Lo += 6.
		CarryToHi = Lo > $0F ? 1 : 0.
		HiSum = ui16(**A**.L >> 4) + ui16(Operand >> 4) + CarryToHi.
		V flag = (~((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum > 9) HiSum += 6.
		C flag = HiSum > $0F.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ADC (72)	Cycles: 5	Size: 2
	Direct Indirect (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Pointer:**DB**	Store as Operand.

6.1		If **D** flag == 0:
		Result = ui16(**A**.L) + ui16(Operand) + **C**.
		V flag = (~(ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result > $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = ui16(**A**.L & 0x0F) + ui16(Operand & 0x0F) + **C**;
		if (Lo > 9) Lo += 6.
		CarryToHi = Lo > $0F ? 1 : 0.
		HiSum = ui16(**A**.L >> 4) + ui16(Operand >> 4) + CarryToHi.
		V flag = (~((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum > 9) HiSum += 6.
		C flag = HiSum > $0F.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ADC (73)	Cycles: 7	Size: 2
	Stack Relative Indirect Indexed (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**. Set Address to Pointer + (**S**.L | $0100).
2.2		
3.1		
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). Bank = **DB** + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.

7.1		If **D** flag == 0:
		Result = ui16(**A**.L) + ui16(Operand) + **C**.
		V flag = (~(ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result > $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = ui16(**A**.L & 0x0F) + ui16(Operand & 0x0F) + **C**;
		if (Lo > 9) Lo += 6.
		CarryToHi = Lo > $0F ? 1 : 0.
		HiSum = ui16(**A**.L >> 4) + ui16(Operand >> 4) + CarryToHi.
		V flag = (~((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum > 9) HiSum += 6.
		C flag = HiSum > $0F.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STZ (74)	Cycles: 4	Size: 2
	Direct, X (Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		If **D**.L is 0, Address.L = Operand + **X**.L + **D**.L, Address.H = **D**.H, otherwise Address = Operand + **X** + **D**. If **D**.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Write to Address	Write 0.
5.1		
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ADC (75)	Cycles: 4	Size: 2
	Direct, X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		If **D**.L is 0, Address.L = Operand + **X**.L + **D**.L, Address.H = **D**.H, otherwise Address = Operand + **X** + **D**. If **D**.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.

5.1		If **D** flag == 0:
		Result = ui16(**A**.L) + ui16(Operand) + **C**.
		V flag = (~(ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result > $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = ui16(**A**.L & 0x0F) + ui16(Operand & 0x0F) + **C**;
		if (Lo > 9) Lo += 6.
		CarryToHi = Lo > $0F ? 1 : 0.
		HiSum = ui16(**A**.L >> 4) + ui16(Operand >> 4) + CarryToHi.
		V flag = (~((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum > 9) HiSum += 6.
		C flag = HiSum > $0F.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ROR (76)	Cycles: 6	Size: 2
	Direct, X (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		If **D**.L is 0, Address.L = Operand + **X**.L + **D**.L, Address.H = **D**.H, otherwise Address = Operand + **X** + **D**. If **D**.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1		
5.2	Write to Address	Write Operand.L.
6.1		Set **C** based off (Operand.L & $01). Perform Operand.L = (Operand.L >> 1) | (**C** << 7). Set **N** based off (Operand.L & $80) and **Z** based off Operand.L.
6.2	Write to Address	Write Operand.L.
7.1		
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ADC (77)	Cycles: 6	Size: 2
	Direct Indirect Indexed Long [d],y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	If **D**.l is not 0, read (Address + 1), otherwise read (Address & $FF00) | ui8(Address.L + 1)	Store as Pointer.H.
5.1		
5.2	If **D**.l is not 0, read (Address + 2), otherwise read (Address & $FF00) | ui8(Address.L + 2)	Store as Bank.
6.1		Perform Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). Bank += (Tmp >> 16).
6.2	Read Pointer:Bank	Store as Operand.

7.1		If **D** flag == 0:
		Result = ui16(**A**.L) + ui16(Operand) + **C**.
		V flag = (~(ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result > $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = ui16(**A**.L & 0x0F) + ui16(Operand & 0x0F) + **C**;
		if (Lo > 9) Lo += 6.
		CarryToHi = Lo > $0F ? 1 : 0.
		HiSum = ui16(**A**.L >> 4) + ui16(Operand >> 4) + CarryToHi.
		V flag = (~((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum > 9) HiSum += 6.
		C flag = HiSum > $0F.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

SEI (78)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Set the **I** flag to 1.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ADC (79)	Cycles: 4	Size: 3
	Absolute, Y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**. Perform Orig = Pointer. Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = **DB** + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.

5.1		If **D** flag == 0:
		Result = ui16(**A**.L) + ui16(Operand) + **C**.
		V flag = (~(ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result > $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = ui16(**A**.L & 0x0F) + ui16(Operand & 0x0F) + **C**;
		if (Lo > 9) Lo += 6.
		CarryToHi = Lo > $0F ? 1 : 0.
		HiSum = ui16(**A**.L >> 4) + ui16(Operand >> 4) + CarryToHi.
		V flag = (~((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum > 9) HiSum += 6.
		C flag = HiSum > $0F.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

PLY (7A)	Cycles: 4	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		
2.2		
3.1		Inc. **S**.L by 1 and set **S**.H to 1.
3.2	Read u8(**S**.L+0) | $0100	Store as Operand.L.
4.1	Set **Y**.L to Operand.L. Set **N** based off (**Y**.L & $80). Set **Z** based off **Y**.L.	
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

TDC (7B)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Set **A** to **D**. Set **N** based off (**A**.H & $80) and **Z** based off **A**.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

JMP (7C)	Cycles: 6	Size: 3
	Absolute Indexed Indirect (a,x) (Jump	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Perform Pointer += **X**. Bank = **PB**.
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Address.L.
5.1		
5.2	Read Pointer + 1:Bank	Store as Address.H.
6.1		Set **PC** to Address.
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ADC (7D)	Cycles: 4	Size: 3
	Absolute, X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. PC. Perform Orig = Pointer. Tmp = ui32(Pointer + **X**). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = **DB** + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.

5.1		If **D** flag == 0:
		Result = ui16(**A**.L) + ui16(Operand) + **C**.
		V flag = (~(ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result > $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = ui16(**A**.L & 0x0F) + ui16(Operand & 0x0F) + **C**;
		if (Lo > 9) Lo += 6.
		CarryToHi = Lo > $0F ? 1 : 0.
		HiSum = ui16(**A**.L >> 4) + ui16(Operand >> 4) + CarryToHi.
		V flag = (~((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum > 9) HiSum += 6.
		C flag = HiSum > $0F.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ROR (7E)	Cycles: 7	Size: 3
	Absolute, X (Read/Modiy/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**. Perform Tmp = ui32(Pointer + **X**). Pointer = ui16(Tmp). Bank = **DB** + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		
5.2	Write to Pointer:Bank	Write Operand.L.
6.1		Set **C** based off (Operand.L & $01). Perform Operand.L = (Operand.L >> 1) | (**C** << 7). Set **N** based off (Operand.L & $80) and **Z** based off Operand.L.
6.2	Write to Pointer:Bank	Write Operand.L.
7.1		
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

ADC (7F)	Cycles: 5	Size: 4
	Absolute Long, X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**.
3.2	Read **PC**:**PB**	Stores as Bank.
4.1		Inc. **PC**. Perform Tmp = ui32(Pointer + **X**). Pointer = ui16(Tmp). Bank += (Tmp >> 16).
4.2	Read Pointer:Bank	Store as Operand.

5.1		If **D** flag == 0:
		Result = ui16(**A**.L) + ui16(Operand) + **C**.
		V flag = (~(ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result > $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = ui16(**A**.L & 0x0F) + ui16(Operand & 0x0F) + **C**;
		if (Lo > 9) Lo += 6.
		CarryToHi = Lo > $0F ? 1 : 0.
		HiSum = ui16(**A**.L >> 4) + ui16(Operand >> 4) + CarryToHi.
		V flag = (~((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum > 9) HiSum += 6.
		C flag = HiSum > $0F.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

BRA (80)	Cycles: 3	Size: 2
	Relative	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**. Decide to branch always.
1.2	Read **PC**:**PB**	Store as Operand. Address = i16(i8(Operand.L)) + **PC**, BoundaryCrossed = Address.H != **PC**.H. If not branching or not BoundaryCrossed, poll interrupts.
2.1		Inc. **PC**. If not branching, end (next half-cycle is 4.2).
2.2		
3.1		Set **PC**.L to Address.L. If not BoundaryCrossed, end (next half-cycle is 4.2).
3.2		Poll interrupts.
4.1		Set **PC**.H to Address.H
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STA (81)	Cycles: 6	Size: 2
	Direct Indexed Indirect (d,x) (Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**. Set Pointer to Pointer + **X** + **D**.
		Address.L = Pointer.L.
		Address.H = **D**.H.
2.2		If **D**.L is 0, skip the next cycle.
3.1		Address.H = Pointer.H (fixes high byte of address).
3.2		
4.1		
4.2	Read Address	Store as Pointer.
5.1		
5.2	* If **D**.l is not 0, read (Address + 1), otherwise read (Address & $FF00) | ui8(Address.L + 1)	Store as Pointer.H.
6.1		Set Operand to A.
6.2	Write to Pointer:**DB**	Write Operand.L.
7.1		
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

* The Single-Step Tests require this to be “Read Address + 1,” but this is an error in the tests.

BRL (82)	Cycles: 4	Size: 3
	Program Counter Relative Long	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Operand.H.
3.1		Inc. **PC**.
3.2		
4.1		Perform **PC** += Operand.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STA (83)	Cycles: 4	Size: 2
	Stack Relative (Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**. Set Address to Pointer + (**S**.L | $0100).
2.2		
3.1		Set Operand to **A**.
3.2	Write to Address	Write Operand.L.
4.1		
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STY (84)	Cycles: 3	Size: 2
	Direct (Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Write to Address	Write **Y**.L.
4.1		
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STA (85)	Cycles: 3	Size: 2
	Direct (Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Write to Address	Write **A**.L.
4.1		
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STX (86)	Cycles: 3	Size: 2
	Direct (Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Write to Address	Write **X**.L.
4.1		
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STA (87)	Cycles: 6	Size: 2
	Direct Indirect Long (Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Address + 2	Store as Bank.
6.1		Set Operand to **A**.
6.2	Write to Pointer:Bank	Write Operand.L.
7.1		
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

DEY (88)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Perform **Y**.L -= 1, set **N** based off (**Y**.L & $80), set **Z** based off **Y**.L.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

BIT (89)	Cycles: 2	Size: 2
	Immediate	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**. Set **Z** based off (**A**.L & Operand.L).
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

TXA (8A)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Perform **A**.L = **X**.L, set **N** based off (**A**.L & $80), set **Z** based off **A**.L.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

PHB (8B)	Cycles: 3	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		
2.2	Write to ((**S**.L | $0100)+0)	Push **DB** onto stack.
3.1		Dec. **S**.L by 1 and set **S**.H to 1.
3.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STY (8C)	Cycles: 4	Size: 3
	Absolute (Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc **PC**. Set Operand to **Y**.
3.2	Write to Address:**DB**	Write Operand.L.
4.1		
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STA (8D)	Cycles: 4	Size: 3
	Absolute (Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc **PC**. Set Operand to **A**.
3.2	Write to Address:**DB**	Write Operand.L.
4.1		
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STX (8E)	Cycles: 4	Size: 3
	Absolute (Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc **PC**. Set Operand to **X**.
3.2	Write to Address:**DB**	Write Operand.L.
4.1		
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STA (8F)	Cycles: 5	Size: 4
	Absolute Long (Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read **PC**:**PB**	Stores as Bank.
4.1		Inc **PC**. Set Operand to **A**.
4.2	Write to Address:Bank	Write Operand.L.
5.1		
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

BCC (90)	Cycles: 2	Size: 2
	Relative	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**. Decide to branch if **C** is 0.
1.2	Read **PC**:**PB**	Store as Operand. Address = i16(i8(Operand.L)) + **PC**, BoundaryCrossed = Address.H != **PC**.H. If not branching or not BoundaryCrossed, poll interrupts.
2.1		Inc. **PC**. If not branching, end (next half-cycle is 4.2).
2.2		
3.1		Set **PC**.L to Address.L. If not BoundaryCrossed, end (next half-cycle is 4.2).
3.2		Poll interrupts.
4.1		Set **PC**.H to Address.H
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STA (91)	Cycles: 6	Size: 2
	Direct Indirect Indexed (d),y (Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). Bank = **DB** + (Tmp >> 16).
5.2		
6.1		Set Operand to **A**.
6.2	Write to Pointer:Bank	Write Operand.L.
7.1		
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STA (92)	Cycles: 5	Size: 2
	Direct Indirect (Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Set Operand to **A**.
5.2	Write to Pointer:**DB**	Write Operand.L.
6.1		
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STA (93)	Cycles: 7	Size: 2
	Stack Relative Indirect Indexed (Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**. Set Address to Pointer + (**S**.L | $0100).
2.2		
3.1		
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). Bank = **DB** + (Tmp >> 16).
5.2		
6.1		Set Operand to **A**.
6.2	Write to Pointer:Bank	Write Operand.L.
7.1		
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STY (94)	Cycles: 4	Size: 2
	Direct, X (Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		If **D**.L is 0, Address.L = Operand + **X**.L + **D**.L, Address.H = **D**.H, otherwise Address = Operand + **X** + **D**. If **D**.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Write to Address	Write **Y**.L.
5.1		
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STA (95)	Cycles: 4	Size: 2
	Direct, X (Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		If **D**.L is 0, Address.L = Operand + **X**.L + **D**.L, Address.H = **D**.H, otherwise Address = Operand + **X** + **D**. If **D**.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Write to Address	Write **A**.L.
5.1		
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STX (96)	Cycles: 4	Size: 2
	Direct, Y (Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		If **D**.L is 0, Address.L = Operand + **Y**.L + **D**.L, Address.H = **D**.H, otherwise Address = Operand + **Y** + **D**. If **D**.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Write to Address	Write **X**.L.
5.1		
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STA (97)	Cycles: 6	Size: 2
	Direct Indirect Indexed Long [d],y (Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	If **D**.l is not 0, read (Address + 1), otherwise read (Address & $FF00) | ui8(Address.L + 1)	Store as Pointer.H.
5.1		
5.2	If **D**.l is not 0, read (Address + 2), otherwise read (Address & $FF00) | ui8(Address.L + 2)	Store as Bank.
6.1		Perform Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). Bank += (Tmp >> 16).
6.2	Write to Pointer:Bank	Write **A**.L.
7.1		
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

TYA (98)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Perform **A**.L = **Y**.L, set **N** based off (**A**.L & $80), set **Z** based off **A**.L.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STA (99)	Cycles: 5	Size: 3
	Absolute, Y (Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**. Perform Tmp = ui32(Pointer + **Y**).
		Pointer = ui16(Tmp).
		Bank = **DB** + (Tmp >> 16).
3.2		
4.1		Set Operand to **A**.
4.2	Write to Pointer:Bank	Write Operand.L.
5.1		
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

TXS (9A)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Set **S**.L to **X**.L and **S**.H to 1
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

TXY (9B)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Perform **Y**.L = **X**.L, set **N** based off (**Y**.L & $80), set **Z** based off **Y**.L.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STZ (9C)	Cycles: 4	Size: 3
	Absolute (Write)	
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc **PC**. Set Operand to 0.
3.2	Write to Address:**DB**	Write Operand.L.
4.1		
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STA (9D)	Cycles: 5	Size: 3
	Absolute, X (Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**. Perform Tmp = ui32(Pointer + **X**).
		Pointer = ui16(Tmp).
		Bank = **DB** + (Tmp >> 16).
3.2		
4.1		Set Operand to **A**.
4.2	Write to Pointer:Bank	Write Operand.L.
5.1		
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STZ (9E)	Cycles: 5	Size: 3
	Absolute, X (Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**. Perform Tmp = ui32(Pointer + **X**).
		Pointer = ui16(Tmp).
		Bank = **DB** + (Tmp >> 16).
3.2		
4.1		Set Operand to 0.
4.2	Write to Pointer:Bank	Write Operand.L.
5.1		
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STA (9F)	Cycles: 5	Size: 4
	Absolute Long, X (Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**.
3.2	Read **PC**:**PB**	Stores as Bank.
4.1		Inc. **PC**. Perform Tmp = ui32(Pointer + **X**).
		Pointer = ui16(Tmp).
		Bank += (Tmp >> 16).
4.2	Write to Pointer:Bank	Write **A**.L.
5.1		
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LDY (A0)	Cycles: 2	Size: 2
	Immediate	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**. Set **Y**.L to Operand.L. Set **N** based off (**Y**.L & $80). Set **Z** based off **Y**.L.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LDA (A1)	Cycles: 6	Size: 2
	Direct Indexed Indirect (d,x) (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**. Set Pointer to Pointer + **X** + **D**. Address.L = Pointer.L. Address.H = **D**.H.
2.2		If **D**.L is 0, skip the next cycle.
3.1		Address.H = Pointer.H (fixes high byte of address).
3.2		
4.1		
4.2	Read Address	Store as Pointer.
5.1		
5.2	* If **D**.l is not 0, read (Address + 1), otherwise read (Address & $FF00) | ui8(Address.L + 1)	Store as Pointer.H.
6.1		
6.2	Read Pointer:**DB**	Store as Operand.
7.1		Set **A**.L to Operand.L. Set **N** based off (**A**.L & $80). Set **Z** based off **A**.L.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

* The Single-Step Tests require this to be “Read Address + 1,” but this is an error in the tests.

LDX (A2)	Cycles: 2	Size: 2
	Immediate	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**. Set **X**.L to Operand.L. Set **N** based off (**X**.L & $80). Set **Z** based off **X**.L.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LDA (A3)	Cycles: 4	Size: 2
	Stack Relative (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**. Set Address to Pointer + (**S**.L | $0100).
2.2		
3.1		
3.2	Read Address	Store as Operand.
4.1		Set **A**.L to Operand.L. Set **N** based off (**A**.L & $80). Set **Z** based off **A**.L.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LDY (A4)	Cycles: 3	Size: 2
	Direct (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Operand.
4.1		Set **Y**.L to Operand.L. Set **N** based off (**Y**.L & $80). Set **Z** based off **Y**.L.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LDA (A5)	Cycles: 3	Size: 2
	Direct (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Operand.
4.1		Set **A**.L to Operand.L. Set **N** based off (**A**.L & $80). Set **Z** based off **A**.L.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LDX (A6)	Cycles: 3	Size: 2
	Direct (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Operand.
4.1		Set **X**.L to Operand.L. Set **N** based off (**X**.L & $80). Set **Z** based off **X**.L.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LDA (A7)	Cycles: 6	Size: 2
	Direct Indirect Long (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Address + 2	Store as Bank.
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Set **A**.L to Operand.L. Set **N** based off (**A**.L & $80). Set **Z** based off **A**.L.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

TAY (A8)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Perform **Y**.L = **A**.L, set **N** based off (**Y**.L & $80), set **Z** based off **Y**.L.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LDA (A9)	Cycles: 2	Size: 2
	Immediate	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**. Set **A**.L to Operand.L. Set **N** based off (**A**.L & $80). Set **Z** based off **A**.L.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

TAX (AA)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Perform **X**.L = **A**.L, set **N** based off (**X**.L & $80), set **Z** based off **X**.L.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

PLB (AB)	Cycles: 4	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		
2.2		
3.1		Inc. **S**.L by 1 and set **S**.H to 1.
3.2	Read ((**S**.L | $0100)+0)	Store as Bank.
4.1		Set **DB** to Bank, set **N** based off (**DB** & $80), and set **Z** based off **DB**.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LDY (AC)	Cycles: 4	Size: 3
	Absolute (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address:**DB**	Store as Operand.
4.1		Set **Y**.L to Operand.L. Set **N** based off (**Y**.L & $80). Set **Z** based off **Y**.L.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LDA (AD)	Cycles: 4	Size: 3
	Absolute (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address:**DB**	Store as Operand.
4.1		Set **A**.L to Operand.L. Set **N** based off (**A**.L & $80). Set **Z** based off **A**.L.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LDX (AE)	Cycles: 4	Size: 3
	Absolute (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address:**DB**	Store as Operand.
4.1		Set **X**.L to Operand.L. Set **N** based off (**X**.L & $80). Set **Z** based off **X**.L.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LDA (AF)	Cycles: 5	Size: 4
	Absolute Long (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read **PC**:**PB**	Stores as Bank.
4.1		Inc. **PC**.
4.2	Read Address:Bank	Store as Operand.
5.1		Set **A**.L to Operand.L. Set **N** based off (**A**.L & $80). Set **Z** based off **A**.L.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

BCS (B0)	Cycles: 2	Size: 2
	Relative	
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. **PC**. Decide to branch if **C** is 1.
1.2	Read **PC**:**PB**	Store as Operand. Address = i16(i8(Operand.L)) + **PC**, BoundaryCrossed = Address.H != **PC**.H. If not branching or not BoundaryCrossed, poll interrupts.
2.1		Inc. **PC**. If not branching, end (next half-cycle is 4.2).
2.2		
3.1		Set **PC**.L to Address.L. If not BoundaryCrossed, end (next half-cycle is 4.2).
3.2		Poll interrupts.
4.1		Set **PC**.H to Address.H
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LDA (B1)	Cycles: 5	Size: 2
	Direct Indirect Indexed (d),y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Orig = Pointer. Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = **DB** + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Set **A**.L to Operand.L. Set **N** based off (**A**.L & $80). Set **Z** based off **A**.L.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LDA (B2)	Cycles: 5	Size: 2
	Direct Indirect (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Pointer:**DB**	Store as Operand.
6.1		Set **A**.L to Operand.L. Set **N** based off (**A**.L & $80). Set **Z** based off **A**.L.
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LDA (B3)	Cycles: 7	Size: 2
	Stack Relative Indirect Indexed (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**. Set Address to Pointer + (**S**.L | $0100).
2.2		
3.1		
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). Bank = **DB** + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Set **A**.L to Operand.L. Set **N** based off (**A**.L & $80). Set **Z** based off **A**.L.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LDY (B4)	Cycles: 4	Size: 2
	Direct, X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		If **D**.L is 0, Address.L = Operand + **X**.L + **D**.L, Address.H = **D**.H, otherwise Address = Operand + **X** + **D**. If **D**.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1		Set **Y**.L to Operand.L. Set **N** based off (**Y**.L & $80). Set **Z** based off **Y**.L.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LDA (B5)	Cycles: 4	Size: 2
	Direct, X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		If **D**.L is 0, Address.L = Operand + **X**.L + **D**.L, Address.H = **D**.H, otherwise Address = Operand + **X** + **D**. If **D**.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1		Set **A**.L to Operand.L. Set **N** based off (**A**.L & $80). Set **Z** based off **A**.L.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LDX (B6)	Cycles: 4	Size: 2
	Direct, Y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		If **D**.L is 0, Address.L = Operand + **Y**.L + **D**.L, Address.H = **D**.H, otherwise Address = Operand + **Y** + **D**. If **D**.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1		Set **X**.L to Operand.L. Set **N** based off (**X**.L & $80). Set **Z** based off **X**.L.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LDA (B7)	Cycles: 6	Size: 2
	Direct Indirect Indexed Long [d],y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	If **D**.l is not 0, read (Address + 1), otherwise read (Address & $FF00) | ui8(Address.L + 1)	Store as Pointer.H.
5.1		
5.2	If **D**.l is not 0, read (Address + 2), otherwise read (Address & $FF00) | ui8(Address.L + 2)	Store as Bank.
6.1		Perform Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). Bank += (Tmp >> 16).
6.2	Read Pointer:Bank	Store as Operand.
7.1		Set **A**.L to Operand.L. Set **N** based off (**A**.L & $80). Set **Z** based off **A**.L.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

CLV (B8)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Set the **V** flag to 0.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LDA (B9)	Cycles: 4	Size: 3
	Absolute, Y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**. Perform Orig = Pointer. Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = **DB** + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Set **A**.L to Operand.L. Set **N** based off (**A**.L & $80). Set **Z** based off **A**.L.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

TSX (BA)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Perform **X**.L = **S**.L, set **N** based off (**X**.L & $80), set **Z** based off **X**.L.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

TYX (BB)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Perform **X**.L = **Y**.L, set **N** based off (**X**.L & $80), set **Z** based off **X**.L.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LDY (BC)	Cycles: 4	Size: 3
	Absolute, X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**. Perform Orig = Pointer.
		Tmp = ui32(Pointer + **X**).
		Pointer = ui16(Tmp).
		If Orig.H == Pointer.H, skip 2 half-cycles.
		Bank = **DB** + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Set **Y**.L to Operand.L. Set **N** based off (**Y**.L & $80). Set **Z** based off **Y**.L.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LDA (BD)	Cycles: 4	Size: 3
	Absolute, X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**. Perform Orig = Pointer.
		Tmp = ui32(Pointer + **X**).
		Pointer = ui16(Tmp).
		If Orig.H == Pointer.H, skip 2 half-cycles.
		Bank = **DB** + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Set **A**.L to Operand.L. Set **N** based off (**A**.L & $80). Set **Z** based off **A**.L.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LDX (BE)	Cycles: 4	Size: 3
	Absolute, Y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**. Perform Orig = Pointer.
		Tmp = ui32(Pointer + **Y**).
		Pointer = ui16(Tmp).
		If Orig.H == Pointer.H, skip 2 half-cycles.
		Bank = **DB** + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Set **X**.L to Operand.L. Set **N** based off (**X**.L & $80). Set **Z** based off **X**.L.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

LDA (BF)	Cycles: 5	Size: 4
	Absolute Long, X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**.
3.2	Read **PC**:**PB**	Stores as Bank.
4.1		Inc. **PC**. Perform Tmp = ui32(Pointer + **X**).
		Pointer = ui16(Tmp).
		Bank += (Tmp >> 16).
4.2	Read Pointer:Bank	Store as Operand.
5.1		Set **A**.L to Operand.L. Set **N** based off (**A**.L & $80). Set **Z** based off **A**.L.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

CPY (C0)	Cycles: 2	Size: 2
	Immediate	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**. Set **C** flag based off (**Y**.L >= Operand.L), set **Z** flag based off (**Y**.L == Operand.L), and set **N** flag based off ((**Y**.L - Operand.L) & $80) != 0.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

CMP (C1)	Cycles: 6	Size: 2
	Direct Indexed Indirect (d,x) (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. **PC**. Set Pointer to Pointer + **X** + **D**. Address.L = Pointer.L. Address.H = **D**.H.
2.2		If **D**.L is 0, skip the next cycle.
3.1		Address.H = Pointer.H (fixes high byte of address).
3.2		
4.1		
4.2	Read Address	Store as Pointer.
5.1		
5.2	* If **D**.l is not 0, read (Address + 1), otherwise read (Address & $FF00) | ui8(Address.L + 1)	Store as Pointer.H.
6.1		
6.2	Read Pointer:**DB**	Store as Operand.
7.1		Set **C** flag based off (**A**.L >= Operand.L), set **Z** flag based off (**A**.L == Operand.L), and set **N** flag based off ((**A**.L - Operand.L) & $80) != 0.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

* The Single-Step Tests require this to be “Read Address + 1,” but this is an error in the tests.

REP (C2)	Cycles: 3	Size: 2
	Immediate	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		**P** &= ~(Operand.L & ~(**X** flag | **M** flag)).
3.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

CMP (C3)	Cycles: 4	Size: 2
	Stack Relative (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**. Set Address to Pointer + (**S**.L | $0100).
2.2		
3.1		
3.2	Read Address	Store as Operand.
4.1		Set **C** flag based off (**A**.L >= Operand.L), set **Z** flag based off (**A**.L == Operand.L), and set **N** flag based off ((**A**.L - Operand.L) & $80) != 0.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

CPY (C4)	Cycles: 3	Size: 2
	Direct (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Operand.
4.1		Set **C** flag based off (**Y**.L >= Operand.L), set **Z** flag based off (**Y**.L == Operand.L), and set **N** flag based off ((**Y**.L - Operand.L) & $80) != 0.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

CMP (C5)	Cycles: 3	Size: 2
	Direct (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Operand.
4.1		Set **C** flag based off (**A**.L >= Operand.L), set Z flag based off (**A**.L == Operand.L), and set **N** flag based off ((**A**.L - Operand.L) & $80) != 0.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

DEC (C6)	Cycles: 5	Size: 2
	Direct (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.L.
5.1		Perform Operand.L -= 1, set **N** based off (Operand.L & $80) and **Z** based off Operand.L.
5.2	Write to Address	Write Operand.L.
6.1		
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

CMP (C7)	Cycles: 6	Size: 2
	Direct Indirect Long (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Address + 2	Store as Bank.
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Set **C** flag based off (**A**.L >= Operand.L), set **Z** flag based off (**A**.L == Operand.L), and set **N** flag based off ((**A**.L - Operand.L) & $80) != 0.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

INY (C8)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Perform **Y**.L += 1, set **N** based off (**Y**.L & $80), set **Z** based off **Y**.L.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

CMP (C9)	Cycles: 2	Size: 2
	Immediate	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**. Set **C** flag based off (**A**.L >= Operand.L), set **Z** flag based off (**A**.L == Operand.L), and set **N** flag based off ((**A**.L - Operand.L) & $80) != 0.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

DEX (CA)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Perform **X**.L -= 1, set **N** based off (**X**.L & $80), set **Z** based off **X**.L.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

WAI (CB)	Cycles: 3	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		
2.2		
3.1		
3.2		Enter waiting state. Repeat this cycle forever.

CPY (CC)	Cycles: 4	Size: 3
	Absolute (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address:**DB**	Store as Operand.
4.1		Set **C** flag based off (**Y**.L >= Operand.L), set **Z** flag based off (**Y**.L == Operand.L), and set **N** flag based off ((**Y**.L - Operand.L) & $80) != 0.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

CMP (CD)	Cycles: 4	Size: 3
	Absolute (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address:**DB**	Store as Operand.
4.1		Set **C** flag based off (**A**.L >= Operand.L), set **Z** flag based off (**A**.L == Operand.L), and set **N** flag based off ((**A**.L - Operand.L) & $80) != 0.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

DEC (CE)	Cycles: 6	Size: 3
	Absolute (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address:**DB**	Store as Operand.
4.1		
4.2	Write to Address:**DB**	Write Operand.L.
5.1		Perform Operand.L -= 1, set **N** based off (Operand.L & $80) and **Z** based off Operand.L.
5.2	Write to Address:**DB**	Write Operand.L.
6.1		
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

CMP (CF)	Cycles: 5	Size: 4
	Absolute Long (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read **PC**:**PB**	Stores as Bank.
4.1		Inc. **PC**.
4.2	Read Address:Bank	Store as Operand.
5.1		Set **C** flag based off (**A**.L >= Operand.L), set **Z** flag based off (**A**.L == Operand.L), and set **N** flag based off ((**A**.L - Operand.L) & $80) != 0.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

BNE (D0)	Cycles: 2	Size: 2
	Relative	
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. **PC**. Decide to branch if **Z** is 0.
1.2	Read **PC**:**PB**	Store as Operand. Address = i16(i8(Operand.L)) + **PC**, BoundaryCrossed = Address.H != **PC**.H. If not branching or not BoundaryCrossed, poll interrupts.
2.1		Inc. **PC**. If not branching, end (next half-cycle is 4.2).
2.2		
3.1		Set **PC**.L to Address.L. If not BoundaryCrossed, end (next half-cycle is 4.2).
3.2		Poll interrupts.
4.1		Set **PC**.H to Address.H
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

CMP (D1)	Cycles: 5	Size: 2
	Direct Indirect Indexed (d),y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Orig = Pointer. Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = **DB** + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Set **C** flag based off (**A**.L >= Operand.L), set **Z** flag based off (**A**.L == Operand.L), and set **N** flag based off ((**A**.L - Operand.L) & $80) != 0.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

CMP (D2)	Cycles: 5	Size: 2
	Direct Indirect (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Pointer:**DB**	Store as Operand.
6.1		Set **C** flag based off (**A**.L >= Operand.L), set **Z** flag based off (**A**.L == Operand.L), and set **N** flag based off ((**A**.L - Operand.L) & $80) != 0.
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

CMP (D3)	Cycles: 7	Size: 2
	Stack Relative Indirect Indexed (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**. Set Address to Pointer + (**S**.L | $0100).
2.2		
3.1		
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). Bank = **DB** + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Set **C** flag based off (**A**.L >= Operand.L), set **Z** flag based off (**A**.L == Operand.L), and set **N** flag based off ((**A**.L - Operand.L) & $80) != 0.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

PEI (D4)	Cycles: 6	Size: 2
	Direct Page Indirect	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Operand.
4.1		
4.2	If **D**.L is 0, Read (((Address.L + 1) & $FF) | (Address.H << 8)), otherwise read Address + 1	Store as Bank.
5.1		
5.2	Write to ((**S**.L | $0100)+0)	Push Operand.H onto stack.
6.1		
6.2	Write to ((**S**.L | $0100)-1)	Push Operand.L onto stack.
7.1		Dec. **S**.L by 2 and set **S**.H to 1.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

CMP (D5)	Cycles: 4	Size: 2
	Direct, X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		If **D**.L is 0, Address.L = Operand + **X**.L + **D**.L, Address.H = **D**.H, otherwise Address = Operand + **X** + **D**. If **D**.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1		Set **C** flag based off (**A**.L >= Operand.L), set **Z** flag based off (**A**.L == Operand.L), and set **N** flag based off ((**A**.L - Operand.L) & $80) != 0.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

DEC (D6)	Cycles: 6	Size: 2
	Direct, X (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		If **D**.L is 0, Address.L = Operand + **X**.L + **D**.L, Address.H = **D**.H, otherwise Address = Operand + **X** + **D**. If **D**.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1		
5.2	Write to Address	Write Operand.L.
6.1		Perform Operand.L -= 1, set **N** based off (Operand.L & $80), and **Z** based off Operand.L.
6.2	Write to Address	Write Operand.L.
7.1		
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

CMP (D7)	Cycles: 6	Size: 2
	Direct Indirect Indexed Long [d],y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	If **D**.l is not 0, read (Address + 1), otherwise read (Address & $FF00) | ui8(Address.L + 1)	Store as Pointer.H.
5.1		
5.2	If **D**.l is not 0, read (Address + 2), otherwise read (Address & $FF00) | ui8(Address.L + 2)	Store as Bank.
6.1		Perform Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). Bank += (Tmp >> 16).
6.2	Read Pointer:Bank	Store as Operand.
7.1		Set **C** flag based off (**A**.L >= Operand.L), set **Z** flag based off (**A**.L == Operand.L), and set **N** flag based off ((**A**.L - Operand.L) & $80) != 0.
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

CLD (D8)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Set the **D** flag to 0.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

CMP (D9)	Cycles: 4	Size: 3
	Absolute, Y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**. Perform Orig = Pointer. Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = **DB** + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Set **C** flag based off (**A**.L >= Operand.L), set **Z** flag based off (**A**.L == Operand.L), and set **N** flag based off ((**A**.L - Operand.L) & $80) != 0.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

PHX (DA)	Cycles: 3	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		
2.2	Write to ((**S**.L | $0100)+0)	Push **X**.L onto stack.
3.1		Dec. **S**.L by 1 and set **S**.H to 1.
3.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

STP (DB)	Cycles: 4	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		
2.2		
3.1		
3.2		
4.1		Stop processor.

JML (DC)	Cycles: 6	Size: 3
	Absolute Indirect Long (Jump)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**.
3.2	Read Pointer	Store as Operand.
4.1		
4.2	Read Pointer + 1	Store as Operand.H.
5.1		
5.2	If **D**.L is 0, Read (((Pointer.L + 2) & $FF) | (Pointer.H << 8)), otherwise read Pointer + 2	Store as Bank.
6.1		Set **PC** to Operand. Set **PB** to Bank.
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

CMP (DD)	Cycles: 4	Size: 3
	Absolute, X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**. Perform Orig = Pointer. Tmp = ui32(Pointer + **X**). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = **DB** + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Set **C** flag based off (**A**.L >= Operand.L), set **Z** flag based off (**A**.L == Operand.L), and set **N** flag based off ((**A**.L - Operand.L) & $80) != 0.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

DEC (DE)	Cycles: 7	Size: 3
	Absolute, X (Read/Modiy/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**. Perform Tmp = ui32(Pointer + **X**). Pointer = ui16(Tmp). Bank = **DB** + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		
5.2	Write to Pointer:Bank	Write Operand.L.
6.1		Perform Operand.L -= 1, set **N** based off (Operand.L & $80), and **Z** based off Operand.L.
6.2	Write to Pointer:Bank	Write Operand.L.
7.1		
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

CMP (DF)	Cycles: 5	Size: 4
	Absolute Long, X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**.
3.2	Read **PC**:**PB**	Stores as Bank.
4.1		Inc. **PC**. Perform Tmp = ui32(Pointer + **X**). Pointer = ui16(Tmp). Bank += (Tmp >> 16).
4.2	Read Pointer:Bank	Store as Operand.
5.1		Set **C** flag based off (**A**.L >= Operand.L), set **Z** flag based off (**A**.L == Operand.L), and set **N** flag based off ((**A**.L - Operand.L) & $80) != 0.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

CPX (E0)	Cycles: 2	Size: 2
	Immediate	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**. Set **C** flag based off (**X**.L >= Operand.L), set **Z** flag based off (**X**.L == Operand.L), and set **N** flag based off ((**X**.L - Operand.L) & $80) != 0.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

SBC (E1)	Cycles: 6	Size: 2
	Direct Indexed Indirect (d,x) (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**. Set Pointer to Pointer + **X** + **D**.
		Address.L = Pointer.L.
		Address.H = **D**.H.
2.2		If **D**.L is 0, skip the next cycle.
3.1		Address.H = Pointer.H (fixes high byte of address).
3.2		
4.1		
4.2	Read Address	Store as Pointer.
5.1		
5.2	* If **D**.l is not 0, read (Address + 1), otherwise read (Address & $FF00) | ui8(Address.L + 1)	Store as Pointer.H.
6.1		
6.2	Read Pointer:**DB**	Store as Operand.

7.1		If **D** flag == 0:
		Result = ui16(**A**.L) - ui16(Operand) - (1 - **C**).
		V flag = ((ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result <= $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = int16(**A**.L & 0x0F) - int16(Operand & 0x0F) - (1 - **C**);
		if (Lo < 0) Lo -= 6.
		BorrowToHi = Lo < 0 ? 1 : 0.
		HiSum = int16(**A**.L >> 4) - int16(Operand >> 4) - BorrowToHi.
		V flag = (((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum < 0) HiSum -= 6.
		C flag = HiSum >= 0.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

* The Single-Step Tests require this to be “Read Address + 1,” but this is an error in the tests.

SEP (E2)	Cycles: 3	Size: 2
	Immediate	
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		
2.2		
3.1		Perform **P** |= (Operand.L & ~(**X** flag | **M** flag)).
3.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

SBC (E3)	Cycles: 4	Size: 2
	Stack Relative (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**. Set Address to Pointer + (**S**.L | $0100).
2.2		
3.1		
3.2	Read Address	Store as Operand.

4.1		If **D** flag == 0:
		Result = ui16(**A**.L) - ui16(Operand) - (1 - **C**).
		V flag = ((ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result <= $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = int16(**A**.L & 0x0F) - int16(Operand & 0x0F) - (1 - **C**);
		if (Lo < 0) Lo -= 6.
		BorrowToHi = Lo < 0 ? 1 : 0.
		HiSum = int16(**A**.L >> 4) - int16(Operand >> 4) - BorrowToHi.
		V flag = (((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum < 0) HiSum -= 6.
		C flag = HiSum >= 0.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

CPX (E4)	Cycles: 3	Size: 2
	Direct (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Operand.
4.1		Set **C** flag based off (**X**.L >= Operand.L), set **Z** flag based off (**X**.L == Operand.L), and set **N** flag based off ((**X**.L - Operand.L) & $80) != 0.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

SBC (E5)	Cycles: 3	Size: 2
	Direct (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Operand.

4.1		If **D** flag == 0:
		Result = ui16(**A**.L) - ui16(Operand) - (1 - **C**).
		V flag = ((ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result <= $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = int16(**A**.L & 0x0F) - int16(Operand & 0x0F) - (1 - **C**);
		if (Lo < 0) Lo -= 6.
		BorrowToHi = Lo < 0 ? 1 : 0.
		HiSum = int16(**A**.L >> 4) - int16(Operand >> 4) - BorrowToHi.
		V flag = (((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum < 0) HiSum -= 6.
		C flag = HiSum >= 0.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

INC (E6)	Cycles: 5	Size: 2
	Direct (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.L.
5.1		Perform Operand.L += 1, set **N** based off (Operand.L & $80), set **Z** based off Operand.L.
5.2	Write to Address	Write Operand.L.
6.1		
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

SBC (E7)	Cycles: 6	Size: 2
	Direct Indirect Long (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Address + 2	Store as Bank.
6.1		
6.2	Read Pointer:Bank	Store as Operand.

7.1		If **D** flag == 0:
		Result = ui16(**A**.L) - ui16(Operand) - (1 - **C**).
		V flag = ((ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result <= $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = int16(**A**.L & 0x0F) - int16(Operand & 0x0F) - (1 - **C**);
		if (Lo < 0) Lo -= 6.
		BorrowToHi = Lo < 0 ? 1 : 0.
		HiSum = int16(**A**.L >> 4) - int16(Operand >> 4) - BorrowToHi.
		V flag = (((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum < 0) HiSum -= 6.
		C flag = HiSum >= 0.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

INX (E8)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Perform **X**.L += 1, set **N** based off (**X**.L & $80), set **Z** based off **X**.L.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

SBC (E9)	Cycles: 2	Size: 2
	Immediate	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. PC.
1.2	Read **PC**:**PB**	Store as Operand.

2.1		If **D** flag == 0:
		Result = ui16(**A**.L) - ui16(Operand) - (1 - **C**).
		V flag = ((ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result <= $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = int16(**A**.L & 0x0F) - int16(Operand & 0x0F) - (1 - **C**);
		if (Lo < 0) Lo -= 6.
		BorrowToHi = Lo < 0 ? 1 : 0.
		HiSum = int16(**A**.L >> 4) - int16(Operand >> 4) - BorrowToHi.
		V flag = (((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum < 0) HiSum -= 6.
		C flag = HiSum >= 0.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

NOP (EA)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

XBA (EB)	Cycles: 3	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		
2.2		
3.1		Swap **A**.L and **A**.H, set **N** based off (**A**.L & $80), and set **Z** based off **A**.L.
3.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

CPX (EC)	Cycles: 4	Size: 3
	Absolute (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address:**DB**	Store as Operand.
4.1		Set **C** flag based off (**X**.L >= Operand.L), set **Z** flag based off (**X**.L == Operand.L), and set **N** flag based off ((**X**.L - Operand.L) & $80) != 0.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

SBC (ED)	Cycles: 4	Size: 3
	Absolute (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address:**DB**	Store as Operand.

4.1		If **D** flag == 0:
		Result = ui16(**A**.L) - ui16(Operand) - (1 - **C**).
		V flag = ((ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result <= $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = int16(**A**.L & 0x0F) - int16(Operand & 0x0F) - (1 - **C**);
		if (Lo < 0) Lo -= 6.
		BorrowToHi = Lo < 0 ? 1 : 0.
		HiSum = int16(**A**.L >> 4) - int16(Operand >> 4) - BorrowToHi.
		V flag = (((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum < 0) HiSum -= 6.
		C flag = HiSum >= 0.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

INC (EE)	Cycles: 6	Size: 3
	Absolute (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address:**DB**	Store as Operand.
4.1		
4.2	Write to Address:**DB**	Write Operand.L.
5.1		Perform Operand.L += 1, set **N** based off (Operand.L & $80), set **Z** based off Operand.L.
5.2	Write to Address:**DB**	Write Operand.L.
6.1		
6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

SBC (EF)	Cycles: 5	Size: 4
	Absolute Long (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Address.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read **PC**:**PB**	Stores as Bank.
4.1		Inc. **PC**.
4.2	Read Address:Bank	Store as Operand.

5.1		If **D** flag == 0:
		Result = ui16(**A**.L) - ui16(Operand) - (1 - **C**).
		V flag = ((ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result <= $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = int16(**A**.L & 0x0F) - int16(Operand & 0x0F) - (1 - **C**);
		if (Lo < 0) Lo -= 6.
		BorrowToHi = Lo < 0 ? 1 : 0.
		HiSum = int16(**A**.L >> 4) - int16(Operand >> 4) - BorrowToHi.
		V flag = (((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum < 0) HiSum -= 6.
		C flag = HiSum >= 0.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

BEQ (F0)	Cycles: 2	Size: 2
	Relative	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**. Decide to branch if **Z** is 1.
1.2	Read **PC**:**PB**	Store as Operand. Address = i16(i8(Operand.L)) + **PC**, BoundaryCrossed = Address.H != **PC**.H. If not branching or not BoundaryCrossed, poll interrupts.
2.1		Inc. **PC**. If not branching, end (next half-cycle is 4.2).
2.2		
3.1		Set **PC**.L to Address.L. If not BoundaryCrossed, end (next half-cycle is 4.2).
3.2		Poll interrupts.
4.1		Set **PC**.H to Address.H
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

SBC (F1)	Cycles: 5	Size: 2
	Direct Indirect Indexed (d),y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Orig = Pointer. Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = **DB** + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.

7.1		If **D** flag == 0:
		Result = ui16(**A**.L) - ui16(Operand) - (1 - **C**).
		V flag = ((ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result <= $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = int16(**A**.L & 0x0F) - int16(Operand & 0x0F) - (1 - **C**);
		if (Lo < 0) Lo -= 6.
		BorrowToHi = Lo < 0 ? 1 : 0.
		HiSum = int16(**A**.L >> 4) - int16(Operand >> 4) - BorrowToHi.
		V flag = (((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum < 0) HiSum -= 6.
		C flag = HiSum >= 0.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

SBC (F2)	Cycles: 5	Size: 2
	Direct Indirect (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Pointer:**DB**	Store as Operand.

6.1		If **D** flag == 0:
		Result = ui16(**A**.L) - ui16(Operand) - (1 - **C**).
		V flag = ((ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result <= $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = int16(**A**.L & 0x0F) - int16(Operand & 0x0F) - (1 - **C**);
		if (Lo < 0) Lo -= 6.
		BorrowToHi = Lo < 0 ? 1 : 0.
		HiSum = int16(**A**.L >> 4) - int16(Operand >> 4) - BorrowToHi.
		V flag = (((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum < 0) HiSum -= 6.
		C flag = HiSum >= 0.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

6.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

SBC (F3)	Cycles: 7	Size: 2
	Stack Relative Indirect Indexed	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**. Set Address to Pointer + (**S**.L | $0100).
2.2		
3.1		
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). Bank = **DB** + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.

7.1		If **D** flag == 0:
		Result = ui16(**A**.L) - ui16(Operand) - (1 - **C**).
		V flag = ((ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result <= $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = int16(**A**.L & 0x0F) - int16(Operand & 0x0F) - (1 - **C**);
		if (Lo < 0) Lo -= 6.
		BorrowToHi = Lo < 0 ? 1 : 0.
		HiSum = int16(**A**.L >> 4) - int16(Operand >> 4) - BorrowToHi.
		V flag = (((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum < 0) HiSum -= 6.
		C flag = HiSum >= 0.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

PEA (F4)	Cycles: 5	Size: 3
	Absolute (Push)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Operand.H.
3.1		Inc. **PC**.
3.2	Write to ((**S**.L | $0100)+0)	Push Operand.H onto stack.
4.1		
4.2	Write to ((**S**.L | $0100)-1)	Push Operand.L onto stack.
5.1		Dec. **S**.L by 2 and set **S**.H to 1.
5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

SBC (F5)	Cycles: 4	Size: 2
	Direct, X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		If **D**.L is 0, Address.L = Operand + **X**.L + **D**.L, Address.H = **D**.H, otherwise Address = Operand + **X** + **D**. If **D**.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.

5.1		If **D** flag == 0:
		Result = ui16(**A**.L) - ui16(Operand) - (1 - **C**).
		V flag = ((ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result <= $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = int16(**A**.L & 0x0F) - int16(Operand & 0x0F) - (1 - **C**);
		if (Lo < 0) Lo -= 6.
		BorrowToHi = Lo < 0 ? 1 : 0.
		HiSum = int16(**A**.L >> 4) - int16(Operand >> 4) - BorrowToHi.
		V flag = (((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum < 0) HiSum -= 6.
		C flag = HiSum >= 0.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

INC (F6)	Cycles: 6	Size: 2
	Direct, X (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		If **D**.L is 0, Address.L = Operand + **X**.L + **D**.L, Address.H = **D**.H, otherwise Address = Operand + **X** + **D**. If **D**.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1		
5.2	Write to Address	Write Operand.L.
6.1		Perform Operand.L += 1, set **N** based off (Operand.L & $80), set **Z** based off Operand.L.
6.2	Write to Address	Write Operand.L.
7.1		
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

SBC (F7)	Cycles: 6	Size: 2
	Direct Indirect Indexed Long [d],y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Operand. If **D**.L is 0, skip the next cycle.
2.1		Inc. **PC**.
2.2		
3.1		Inc. **PC** if previous cycle was skipped. Set Address to Operand + **D**.
3.2	Read Address	Store as Pointer.
4.1		
4.2	If **D**.l is not 0, read (Address + 1), otherwise read (Address & $FF00) | ui8(Address.L + 1)	Store as Pointer.H.
5.1		
5.2	If **D**.l is not 0, read (Address + 2), otherwise read (Address & $FF00) | ui8(Address.L + 2)	Store as Bank.
6.1		Perform Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). Bank += (Tmp >> 16).
6.2	Read Pointer:Bank	Store as Operand.

7.1		If **D** flag == 0:
		Result = ui16(**A**.L) - ui16(Operand) - (1 - **C**).
		V flag = ((ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result <= $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = int16(**A**.L & 0x0F) - int16(Operand & 0x0F) - (1 - **C**);
		if (Lo < 0) Lo -= 6.
		BorrowToHi = Lo < 0 ? 1 : 0.
		HiSum = int16(**A**.L >> 4) - int16(Operand >> 4) - BorrowToHi.
		V flag = (((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum < 0) HiSum -= 6.
		C flag = HiSum >= 0.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

SED (F8)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Set the **D** flag to 1.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

SBC (F9)	Cycles: 4	Size: 3
	Absolute, Y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. PC. Perform Orig = Pointer. Tmp = ui32(Pointer + **Y**). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = **DB** + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.

5.1		If **D** flag == 0:
		Result = ui16(**A**.L) - ui16(Operand) - (1 - **C**).
		V flag = ((ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result <= $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = int16(**A**.L & 0x0F) - int16(Operand & 0x0F) - (1 - **C**);
		if (Lo < 0) Lo -= 6.
		BorrowToHi = Lo < 0 ? 1 : 0.
		HiSum = int16(**A**.L >> 4) - int16(Operand >> 4) - BorrowToHi.
		V flag = (((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum < 0) HiSum -= 6.
		C flag = HiSum >= 0.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

PLX (FA)	Cycles: 4	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		
2.2		
3.1		Inc. **S**.L by 1 and set **S**.H to 1.
3.2	Read u8(**S**.L+0) | $0100	Store as Operand.L.
4.1		Set **X**.L to Operand.L, set **N** based off (**X**.L & $80), and set **Z** based off **X**.L.
4.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

XCE (FB)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Discard.
2.1		Exchanges the Carry and Emulation flags. If Emulation mode is entered, **M** and **X** flags are set to 1, **S**.H is set to 1, and **X**.H and **Y**.H are cleared to 0.
2.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

JSR (FC)	Cycles: 8	Size: 3
	Absolute Indexed Indirect (a,x) (Jump)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Write to u8(**S**.L+0) | $0100	Push **PC**.H onto stack.
3.1		
3.2	Write to u8(**S**.L-1) | $0100	Push **PC**.L onto stack.
4.1		Dec. **S**.L by 2 and set **S**.H to 1.
4.2	Read **PC**:**PB**	Store as Pointer.H.
5.1		Perform Pointer += **X**.
		Bank = **PB**.
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Address.L.
7.1		
7.2	Read Pointer + 1:Bank	Store as Address.H.
8.1		Set **PC** to Address.
8.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

SBC (FD)	Cycles: 4	Size: 3
	Absolute, X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**. Perform Orig = Pointer. Tmp = ui32(Pointer + **X**). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = **DB** + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.

5.1		If **D** flag == 0:
		Result = ui16(**A**.L) - ui16(Operand) - (1 - **C**).
		V flag = ((ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result <= $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = int16(**A**.L & 0x0F) - int16(Operand & 0x0F) - (1 - **C**);
		if (Lo < 0) Lo -= 6.
		BorrowToHi = Lo < 0 ? 1 : 0.
		HiSum = int16(**A**.L >> 4) - int16(Operand >> 4) - BorrowToHi.
		V flag = (((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum < 0) HiSum -= 6.
		C flag = HiSum >= 0.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

INC (FE)	Cycles: 7	Size: 3
	Absolute, X (Read/Modiy/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**. Perform Tmp = ui32(Pointer + **X**). Pointer = ui16(Tmp). Bank = **DB** + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		
5.2	Write to Pointer:Bank	Write Operand.L.
6.1		Perform Operand.L += 1, set **N** based off (Operand.L & $80), set **Z** based off Operand.L.
6.2	Write to Pointer:Bank	Write Operand.L.
7.1		
7.2	Read **PC**:**PB**	Store as OpCode.
+X.1		

SBC (FF)	Cycles: 5	Size: 4
	Absolute Long, X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**:**PB**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**:**PB**	Store as Pointer.
2.1		Inc. **PC**.
2.2	Read **PC**:**PB**	Store as Pointer.H.
3.1		Inc. **PC**.
3.2	Read **PC**:**PB**	Stores as Bank.
4.1		Inc. **PC**. Perform Tmp = ui32(Pointer + **X**). Pointer = ui16(Tmp). Bank += (Tmp >> 16).
4.2	Read Pointer:Bank	Store as Operand.

5.1		If **D** flag == 0:
		Result = ui16(**A**.L) - ui16(Operand) - (1 - **C**).
		V flag = ((ui16(**A**.L) ^ ui16(Operand)) & (ui16(**A**.L) ^ ui8(Result)) & $80) != 0.
		C flag = Result <= $FF.
		Result = ui8(Result). **Z** flag = Result == $00. **N** flag = (Result & $80) != 0.
		If **D** flag == 1:
		Lo = int16(**A**.L & 0x0F) - int16(Operand & 0x0F) - (1 - **C**);
		if (Lo < 0) Lo -= 6.
		BorrowToHi = Lo < 0 ? 1 : 0.
		HiSum = int16(**A**.L >> 4) - int16(Operand >> 4) - BorrowToHi.
		V flag = (((**A**.L >> 4) ^ (Operand >> 4)) & ((**A**.L >> 4) ^ HiSum) & $08) != 0.
		if (HiSum < 0) HiSum -= 6.
		C flag = HiSum >= 0.
		Result = ui8(((HiSum & 0x0F) << 4) | (Lo & $0F)).
		Z flag = Result == $00. **N** flag = (Result & $80) != 0.

5.2	Read **PC**:**PB**	Store as OpCode.
+X.1		